MATH 8090:
Spectral Analysis of Time Series II

Readings: CC08 Chapter 14; BD16 Chapters 4 and 10.1; SS17 Chapters 1.5–1.6, 4.4–4.6, 4.8, and 5.5

Whitney Huang



School of
MATHEMATICAL AND
STATISTICAL SCIENCES
Clemson University

Week 12

Agenda

Review

Southern Oscillation Index Example

Parametric Spectral Estimation

Lagged Regression Models

Review: Nonparametric Spectral Estimation

- Periodogram: $I(\omega_j) = |d(\omega_j)|^2$, where

$$d(\omega_j) = n^{-\frac{1}{2}} \sum_{t=1}^{n} y_t e^{-2\pi i \omega_j t}, \ \omega_j = \frac{j}{n}, \ j = 0, 1, \cdots, n-1$$

  - $\frac{I(\omega_j)}{\frac{1}{2} f(\omega_j)} \overset{\text{approx.}}{\sim} \chi^2_2$, $j = 1, \cdots, m = \frac{n-1}{2} \Rightarrow \mathbb{E}[I(\omega_j)] \approx f(\omega_j)$ (unbiased)
  - But $\text{Var}[I(\omega_j)] \approx f^2(\omega_j)$ (inconsistent because it does not go to 0 as $n \to \infty$)

- Smooth the periodogram
  - Averaged periodogram: $\bar{f}(\omega_j) = \frac{1}{L} \sum_{k=-l}^{l} I(\omega_{j+k}), \quad L = 2l + 1$
  - Smoothed periodogram: $\bar{f}(\omega_j) = \sum_{k=-l}^{l} W_l(k) I(\omega_{j+k})$

- Pointwise CI for $f(\omega_j)$:

$$\frac{\nu \bar{f}(\omega_j)}{\chi^2_\nu(1-\alpha/2)} \leq f(\omega_j) \leq \frac{\nu \bar{f}(\omega_j)}{\chi^2_\nu(\alpha/2)}, \quad \nu = \frac{2}{\sum_{k=-l}^{l} W_l^2(k)}$$

Spectral ANOVA

- For odd $n = 2m + 1$, the inverse transform can be written

$$y_t - \bar{y} = \frac{2}{\sqrt{n}} \sum_{j=1}^{m} \left[d_{\cos}(\omega_j) \cos(2\pi\omega_j t) + d_{\sin}(\omega_j) \sin(2\pi\omega_j t) \right].$$

- Square and sum over t; orthogonality of sines and cosines implies that

$$\begin{aligned} \sum_{t=1}^{n} (y_t - \bar{y})^2 &= 2 \sum_{j=1}^{m} \left[d_{\cos}(\omega_j)^2 + d_{\sin}(\omega_j)^2 \right] \\ &= 2 \sum_{j=1}^{m} I(\omega_j) \end{aligned}$$

We have partitioned $\sum_{t=1}^{n} (y_t - \bar{y})^2$ into $2 \times \sum_{j=1}^{m} I(\omega_j)$. This leads to Spectral ANOVA $\Rightarrow$ total variance = sum of frequency components

Spectral ANOVA (Cont'd)

Source	df	SS	MS
ω_1	2	$2I(\omega_1)$	$I(\omega_1)$
ω_2	2	$2I(\omega_2)$	$I(\omega_2)$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
ω_m	2	$2I(\omega_m)$	$I(\omega_m)$
Total	$2m = n - 1$	$\sum(y_t - \bar{y})^2$	

Toy example:

```
> x <- c(1, 2, 3, 2, 1) - mean(x)
> c1 <- cos(2 * pi * (1:5) * (1 / 5)); s1 <- sin(2 * pi * (1:5) * (1 / 5))
> c2 <- cos(2 * pi * (1:5) * (2 / 5)); s2 <- sin(2 * pi * (1:5) * (2 / 5))
> omega1 <- cbind(c1, s1); omega2 <- cbind(c2, s2)
> anova(lm(x ~ omega1 + omega2))
Warning in anova.lm(lm(x ~ omega1 + omega2)) :
  ANOVA F-tests on an essentially perfect fit are unreliable
Analysis of Variance Table

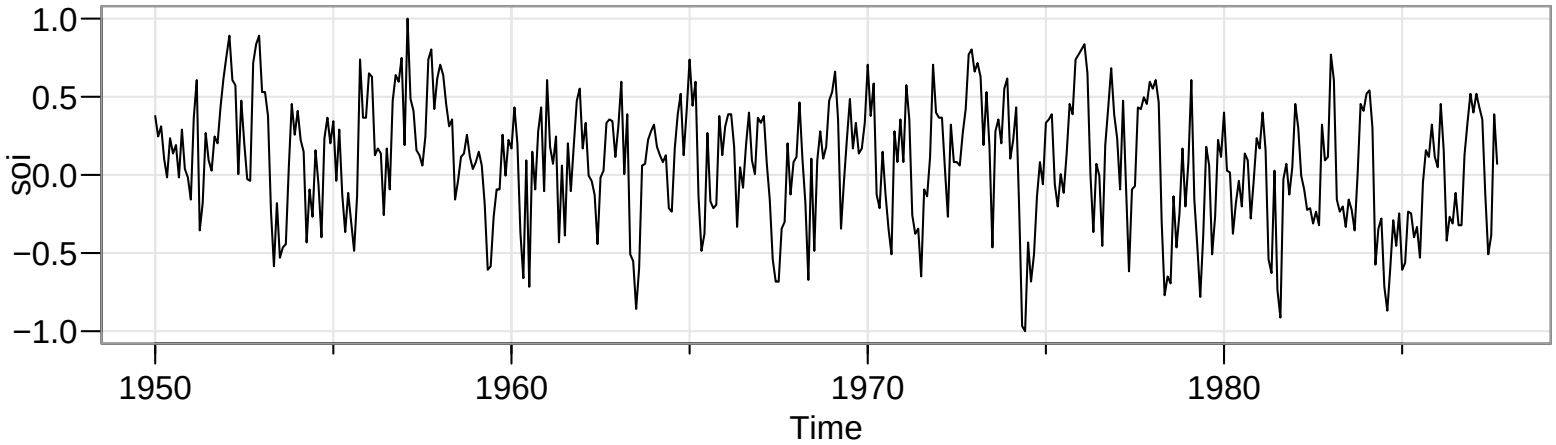
Response: x
          Df  Sum Sq Mean Sq F value Pr(>F)
omega1     2 2.74164 1.37082
omega2     2 0.05836 0.02918
Residuals  0 0.00000
```

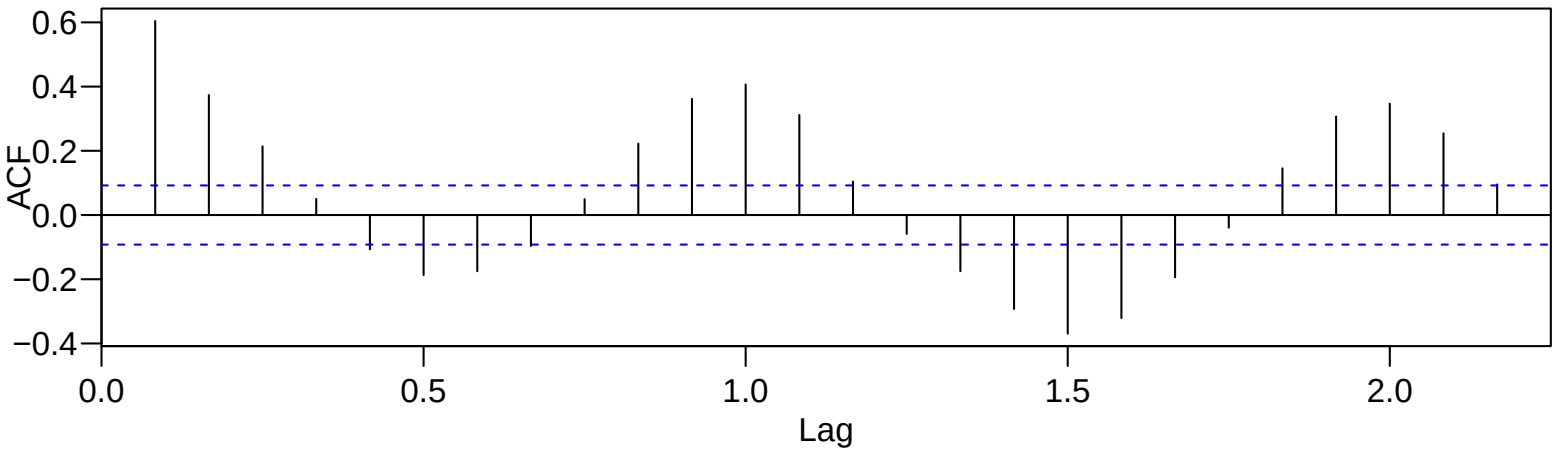
This Week: Three Questions

1. Which frequencies dominate a real time series? SOI and spectral ANOVA
2. Can a fitted ARMA model estimate its spectrum? Parametric spectra
3. How are two series related across lags and frequencies? CCF and coherence

Example: Southern Oscillation Index (SOI)

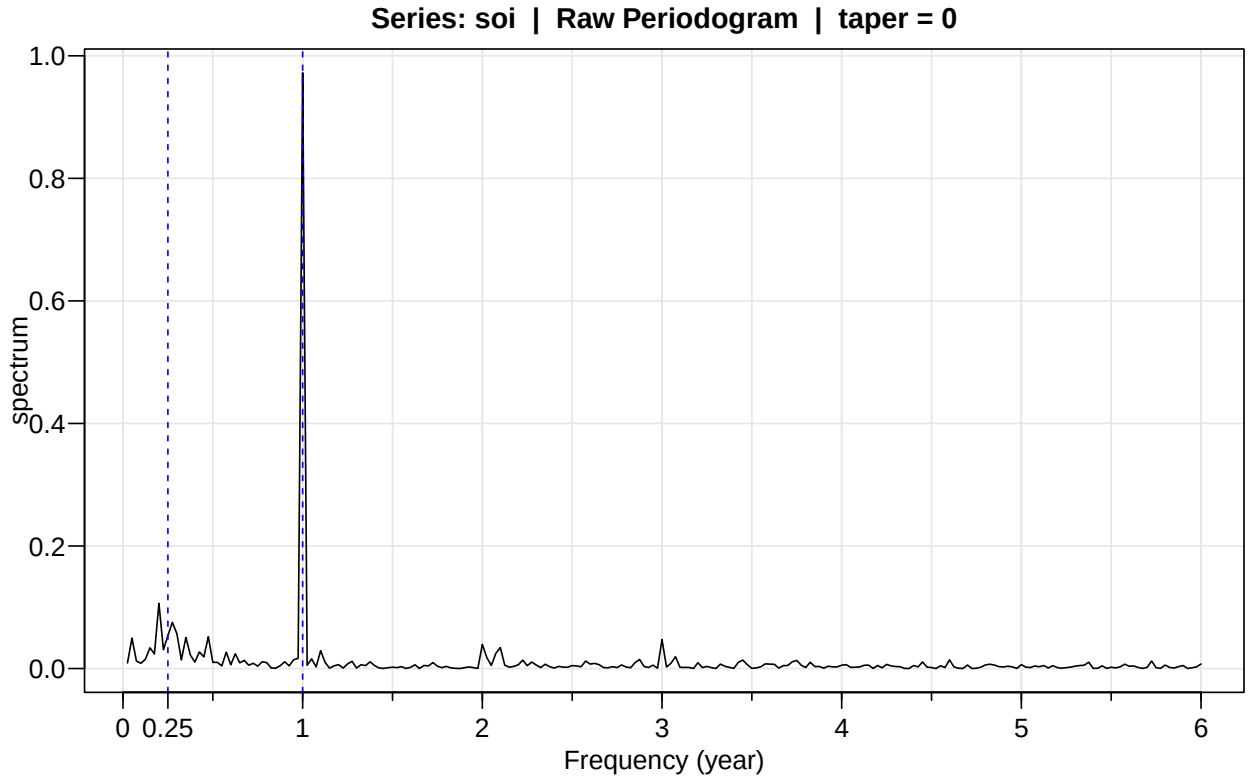
Southern Oscillation Index (SOI) for a period of 453 months ranging over the years 1950-1987





What are the hidden periods of SOI?

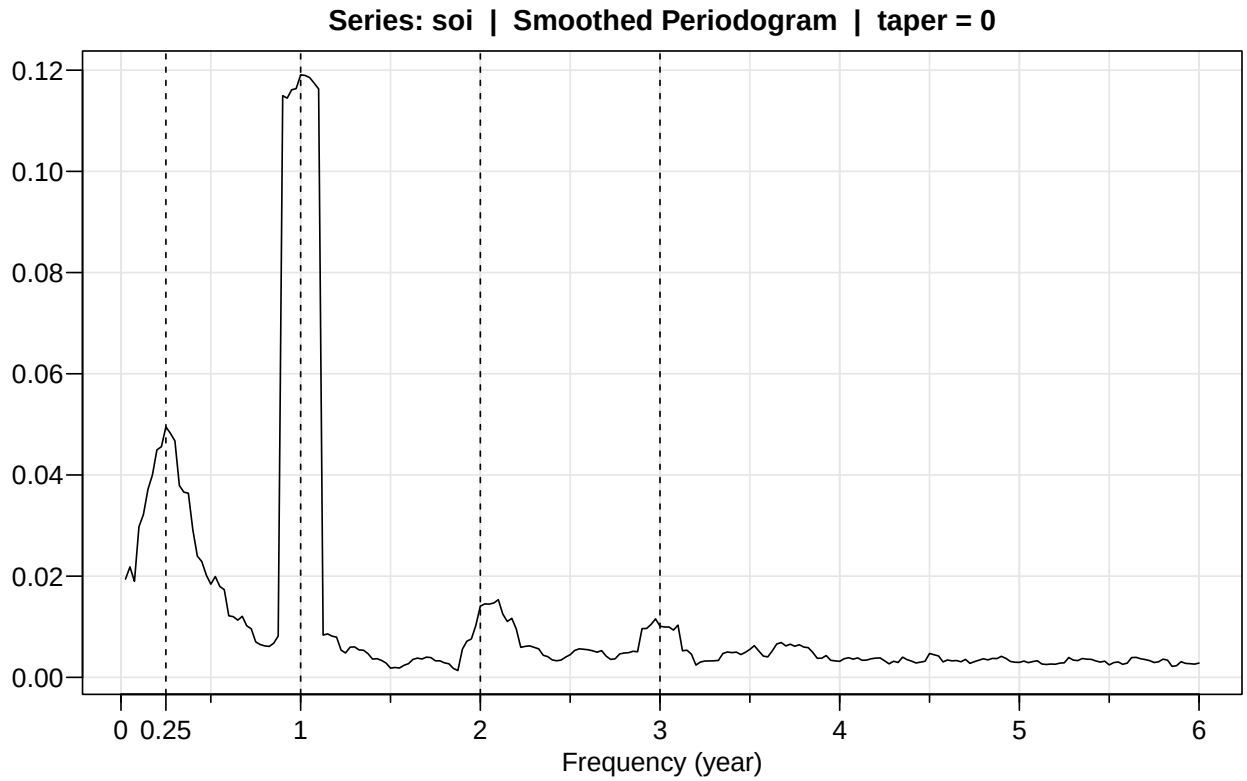
SOI Example: Raw Periodogram



An approximate 95% confidence interval for $f(\omega)$:

ω	Period	Power	Lower	Upper
$\frac{1}{48}$	4 years	0.0537	0.0146	2.1222
$\frac{1}{12}$	1 year	0.9722	0.2636	38.4011

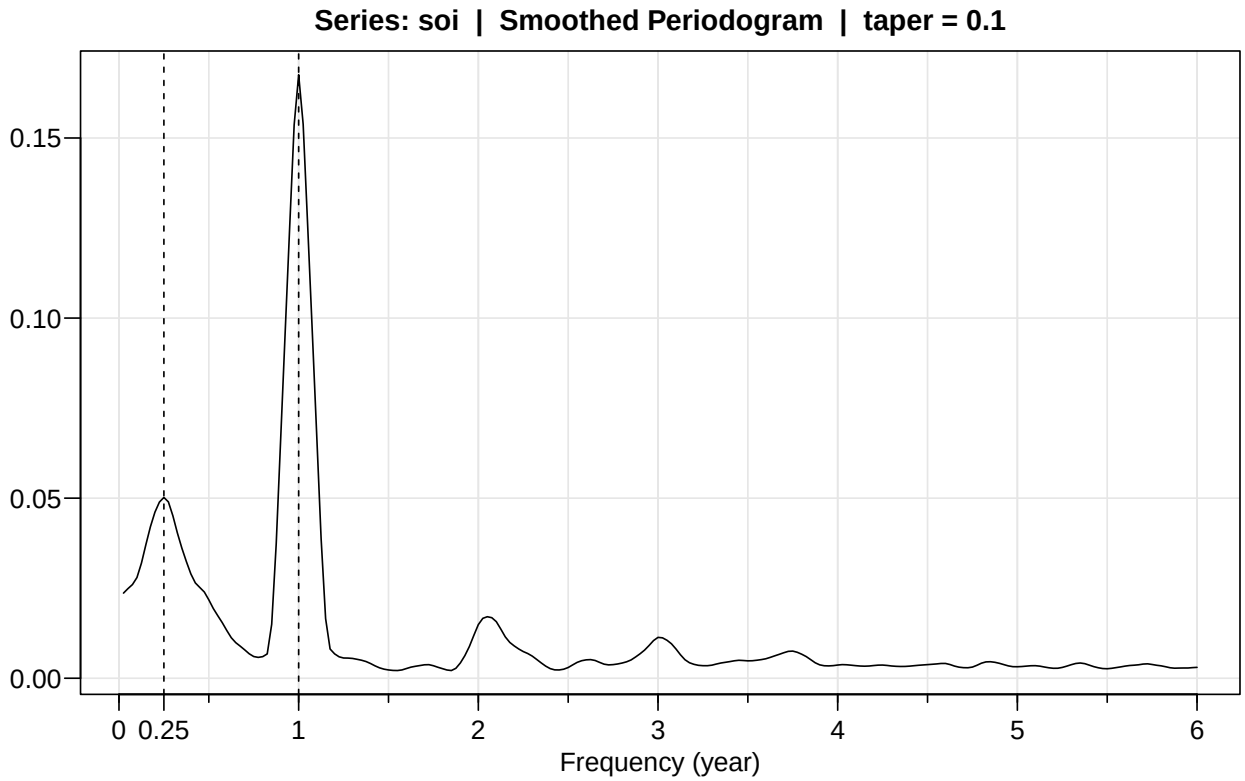
SOI Example: Averaged Periodogram (Daniell with $l = 4$)



An approximate 95% confidence interval for $f(\omega)$:

ω	Period	Power	Lower	Upper
$\frac{1}{48}$	4 years	0.0495	0.0279	0.1113
$\frac{1}{12}$	1 year	0.1191	0.0670	0.2677

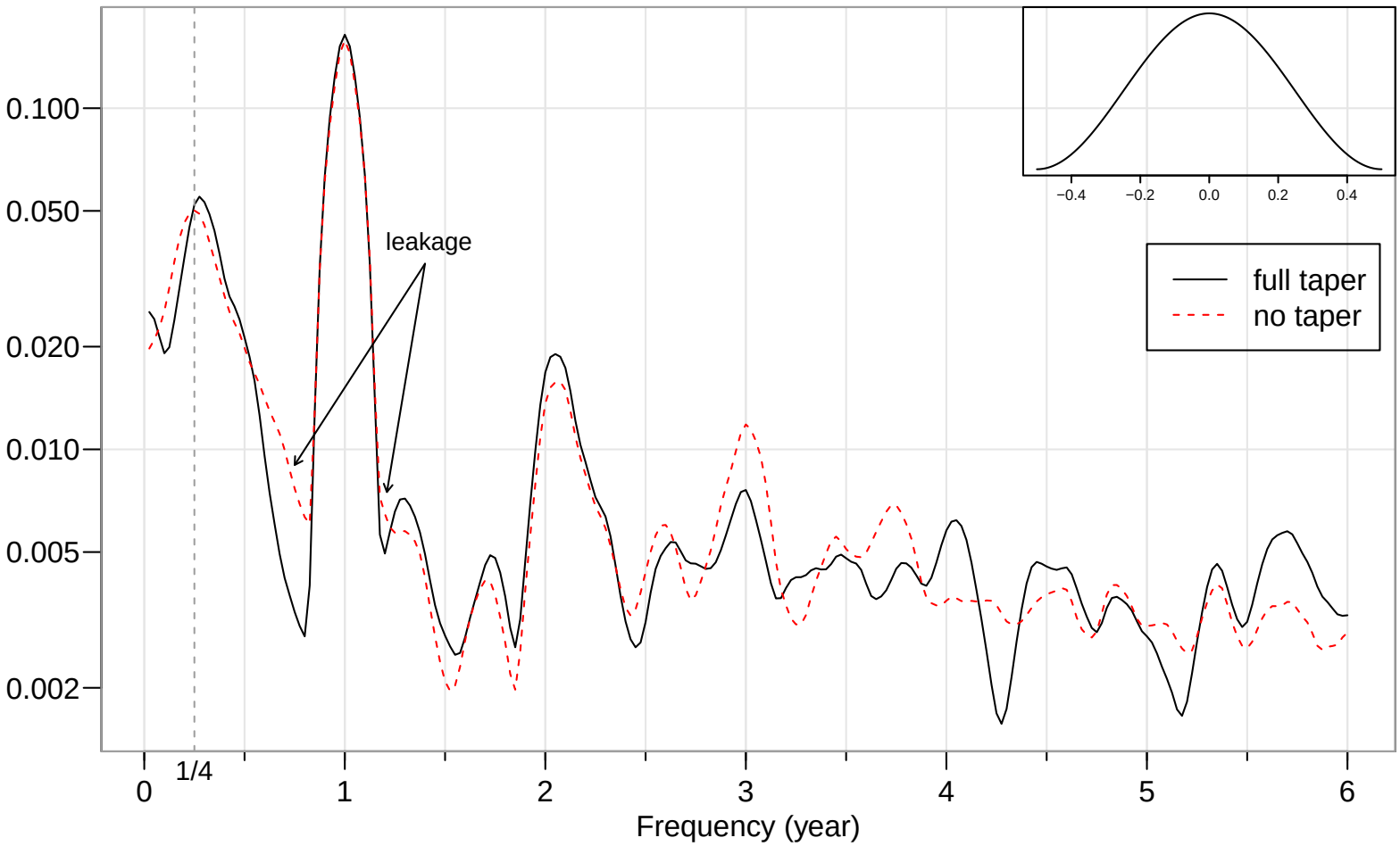
SOI Example: Smoothed Periodogram (modified Daniell $c(3,3)$)



An approximate 95% confidence interval for $f(\omega)$:

ω	Period	Power	Lower	Upper
$\frac{1}{48}$	4 years	0.0502	0.0283	0.1129
$\frac{1}{12}$	1 year	0.1675	0.0943	0.3767

Apply Tapering to Alleviate Spectral Leakage



The tapered spectrum does a better job in separating the annual cycle (period 12 months) and a roughly four-year cycle (period 48 months)

Seasonally Adjusted SOI [**Source**: Peter Bloomfield's ST 730 Lecture Notes]

- The Southern Oscillation Index data provided by Shumway and Stoffer is not seasonally adjusted, which explains the substantial peaks in the periodogram at the annual frequency

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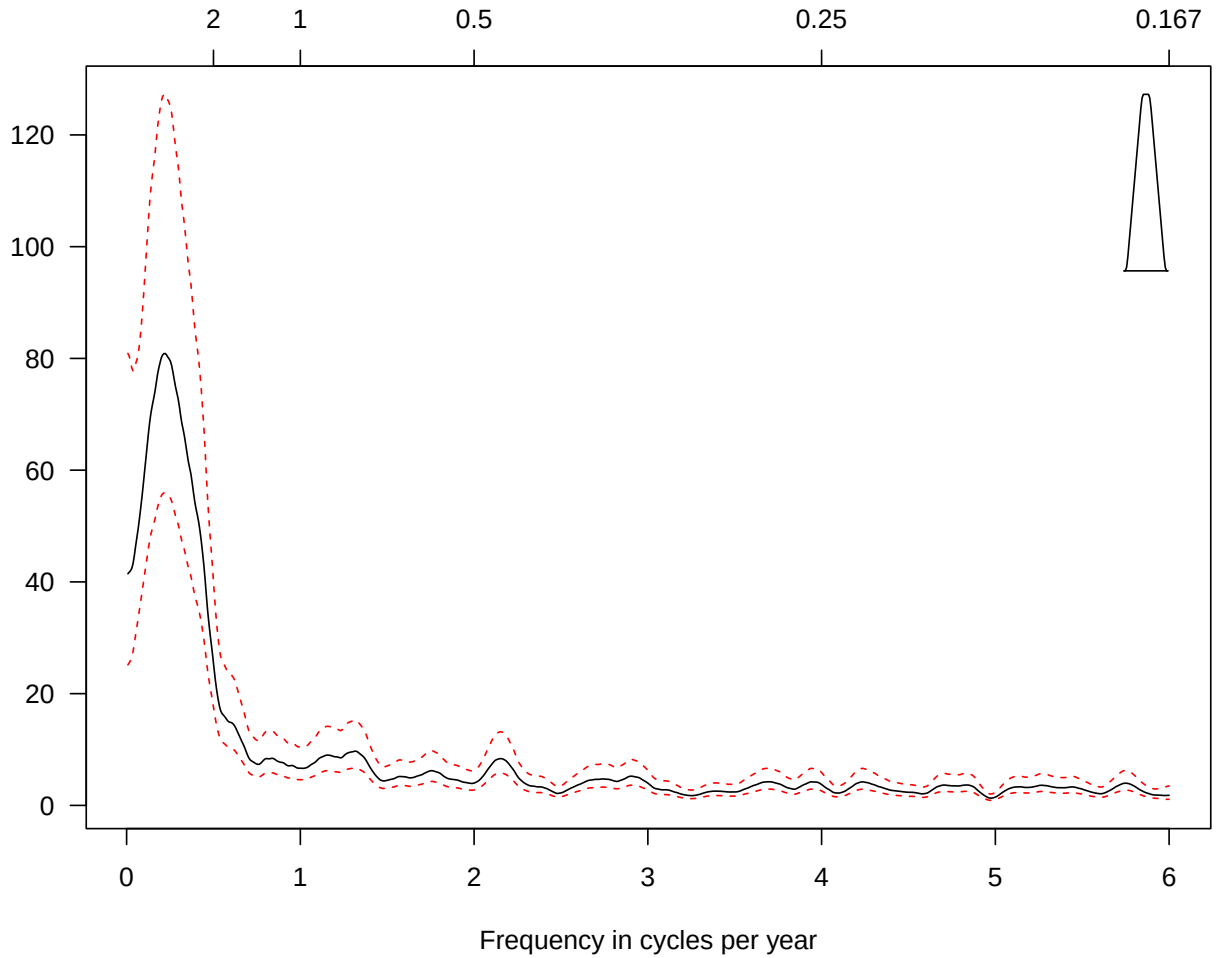
- The Southern Oscillation Index data provided by Shumway and Stoffer is not seasonally adjusted, which explains the substantial peaks in the periodogram at the annual frequency

- A deterministic seasonal mean violates weak stationarity; after removing it, we can study the spectrum of the remaining component if that component is stationary

Seasonally Adjusted SOI [**Source**: Peter Bloomfield's ST 730 Lecture Notes]

- The Southern Oscillation Index data provided by Shumway and Stoffer is not seasonally adjusted, which explains the substantial peaks in the periodogram at the annual frequency
- A deterministic seasonal mean violates weak stationarity; after removing it, we can study the spectrum of the remaining component if that component is stationary
- A more sensible analysis uses the seasonally adjusted series. (Bloomfield did this by fitting a seasonal means model using data from 1876-2010.)

SOI Example from Bloomfield: Smoothed Periodogram



Note that the peak at the annual frequency disappears due to the removal of the annual cycle

Interpreting the SOI Spectral Comparisons

- Raw periodogram: high resolution, but very high variability.
- Daniell smoothing: averages nearby frequencies; peaks become more stable.
- Tapering: mitigates leakage from strong components.
- Seasonal adjustment: removes a deterministic seasonal mean, if present.

A peak near an annual frequency does not, by itself, establish nonstationarity: stationary processes can also exhibit seasonal spectral peaks.

Parametric versus Nonparametric Estimation

- Parametric estimation: estimate a model that is specified by a fixed number of parameters

- Nonparametric estimation: estimate a model that is specified by a number of parameters that can grow as the sample grows

The smoothed periodogram estimates we have considered are **nonparametric**: the estimates of the spectral density can be parameterized by estimated values at ω_j's. As $n \uparrow$, the number of distinct frequency values increases

The time domain models we considered are **parametric**. For example, an ARMA(p,q) process can be completely specified with $p + q + 1$ parameters

Parametric Spectral Estimation

The typical approach is to use the maximum likelihood parameter estimates $(\hat{\phi}_1, \cdots, \hat{\phi}_p, \hat{\sigma}^2)$ for the parameters of an AR(p), and then compute $f(\omega)$ for this estimated AR model:

$$\hat{f}(\omega) = \frac{\hat{\sigma}^2}{|\hat{\phi}(e^{-2\pi i\omega})|^2}$$

For large n,

$\mathrm{Var}(\hat{f}(\omega))$ depends on model, frequency, and estimation method

- Increasing p allows more complex spectral shapes and may reduce approximation bias

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- Increasing p allows more complex spectral shapes and may reduce approximation bias
- Larger p typically increases estimation uncertainty and overfitting risk

ARMA Spectral Estimation

- Sometimes ARMA models are used instead
- Estimate the parameters of an ARMA(p,q) model and compute its spectral density:

$$\hat{f}(\omega) = \hat{\sigma}^2 \left| \frac{\hat{\theta}(e^{-2\pi i\omega})}{\hat{\phi}(e^{-2\pi i\omega})} \right|^2 .$$

- However, it is more common to use large AR models, rather than ARMA models

Parametric versus Nonparametric Spectral Estimation

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- The main advantage of parametric spectral estimation over nonparametric is that it often gives better frequency resolution of a small number of peaks
- This is especially important if there is more than one peak at nearby frequencies
- The disadvantage of parametric spectral estimation is the inflexibility due to the use of the restricted class of ARMA models.

Parametric Spectral Estimation: Summary

Given data $y_1, y_2, \cdots, y_n$,

1. Estimate the AR parameters $(\phi_1, \phi_2, \cdots, \phi_p, \sigma^2)$ using maximum likelihood or Yule-Walker/least squares, choose a suitable model order p using AIC or BIC

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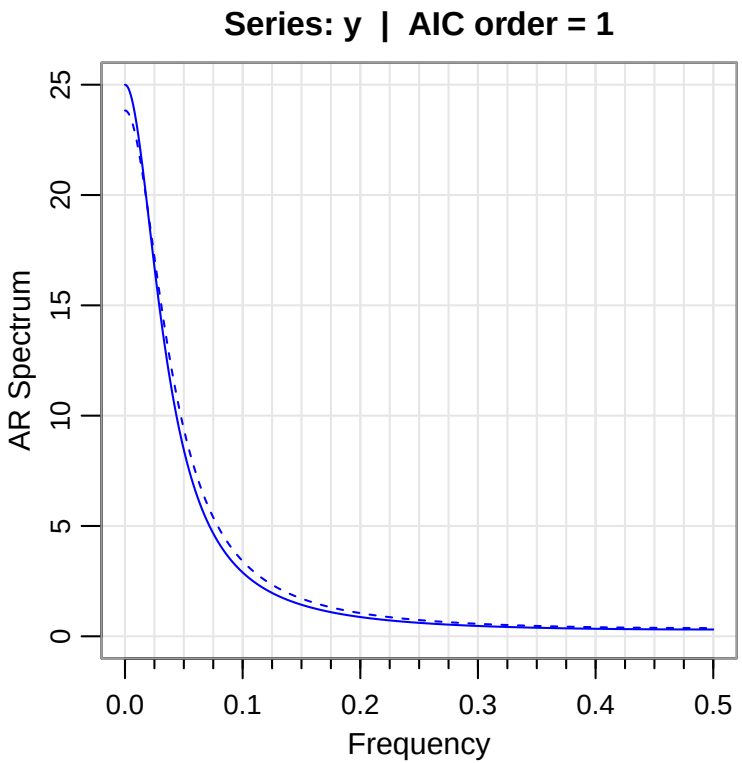
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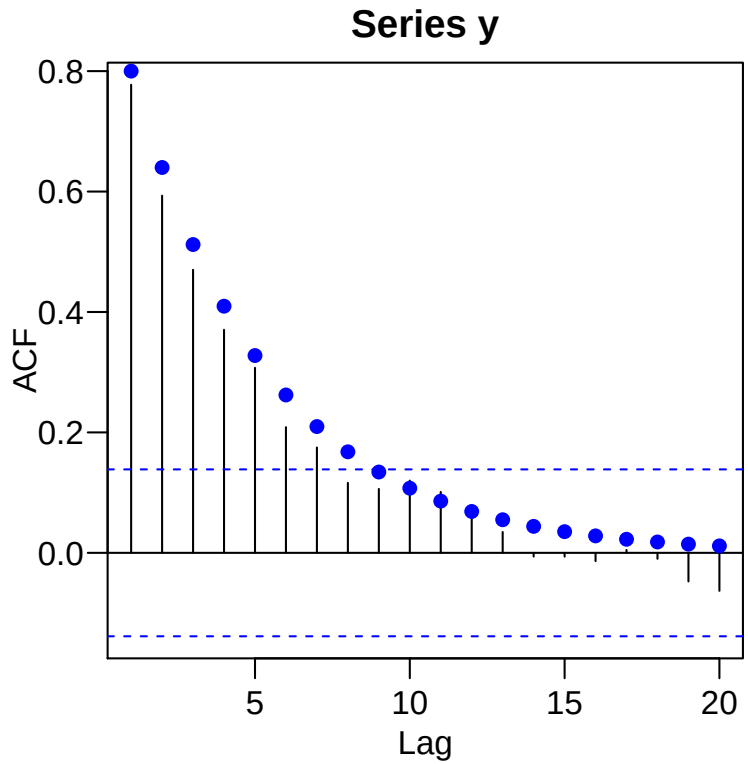
2. Use the estimates $(\hat{\phi}_1, \hat{\phi}_2, \cdots, \hat{\phi}_p, \hat{\sigma}^2)$ to compute the estimated spectral density:

$$\hat{f}(\omega) = \frac{\hat{\sigma}^2}{\left|\hat{\phi}(e^{-2\pi i\omega})\right|^2}$$

Example: AR(1) with $\phi = 0.8$

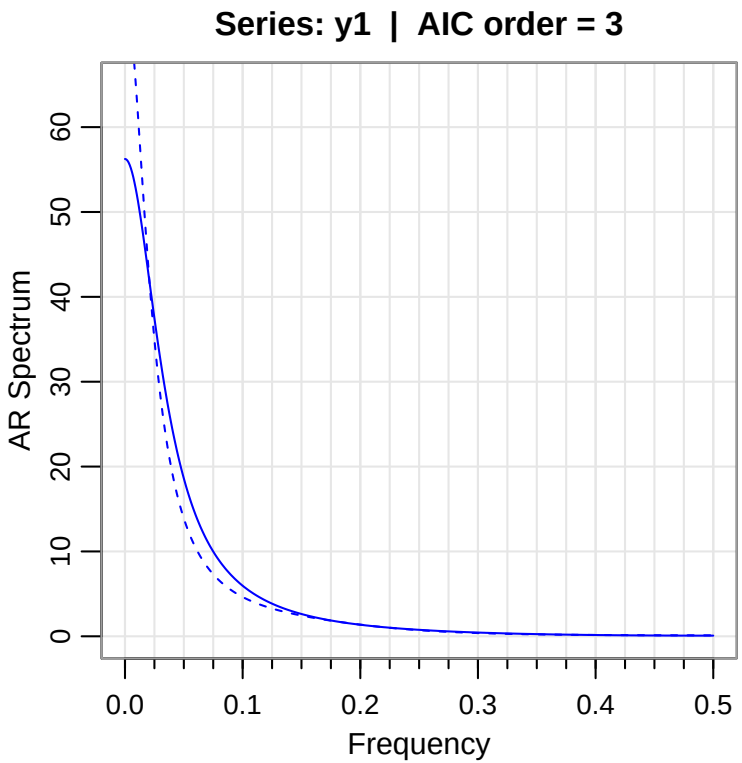
1. Use AIC to select p, the order of the AR model
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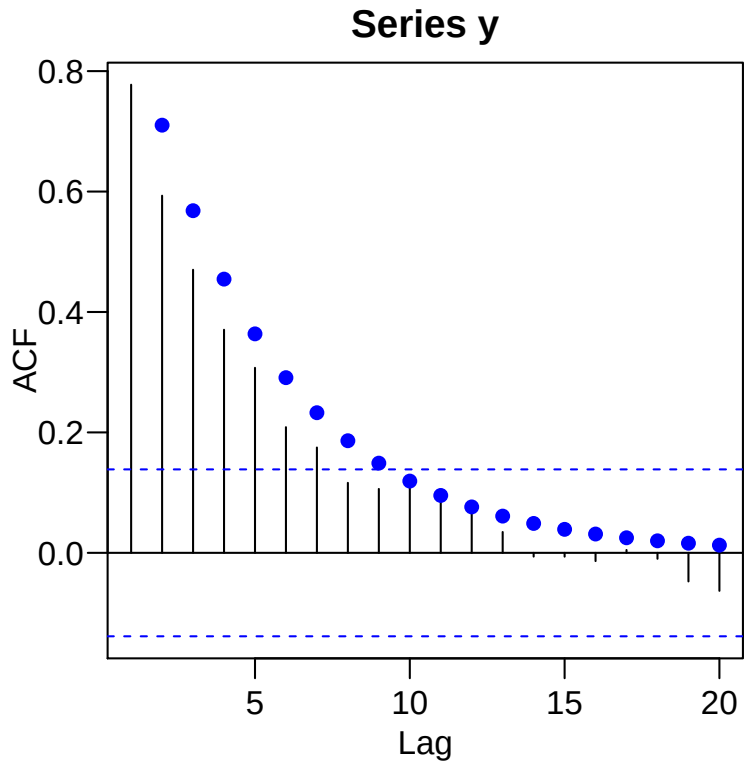




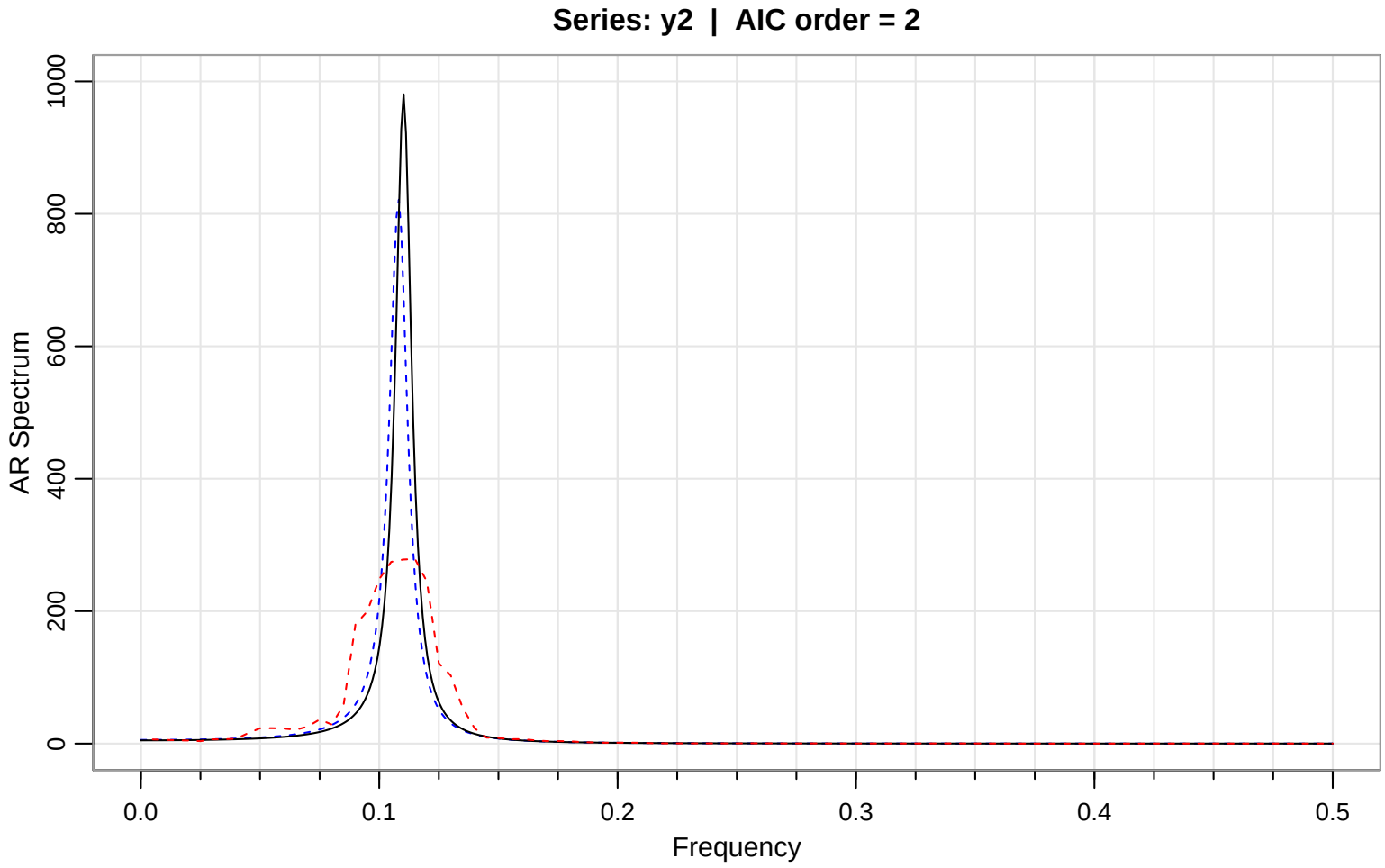
Example: ARMA(1, 1) with $\phi = 0.8$ and $\theta = 0.5$

1. Use AIC to select p, the order of the AR model
2. Use the estimates $(\hat{\phi}_1, \hat{\phi}_2, \cdots, \hat{\phi}_p, \hat{\sigma}^2)$ to compute the estimated spectral



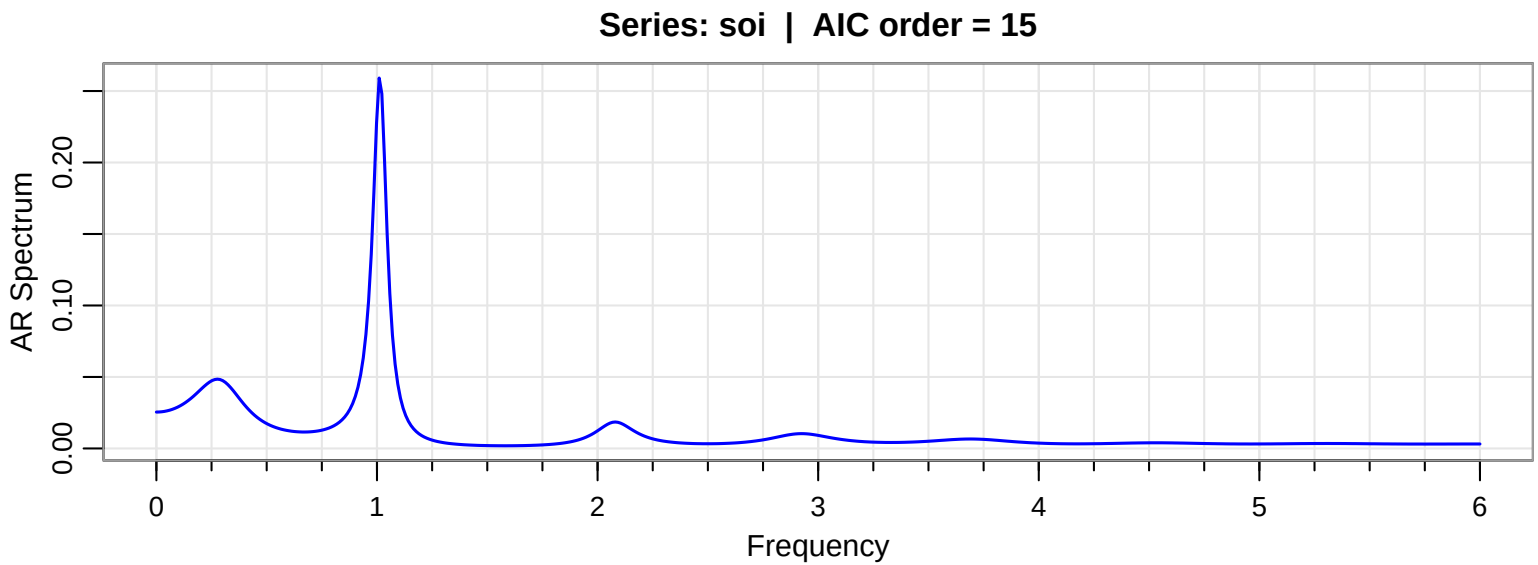


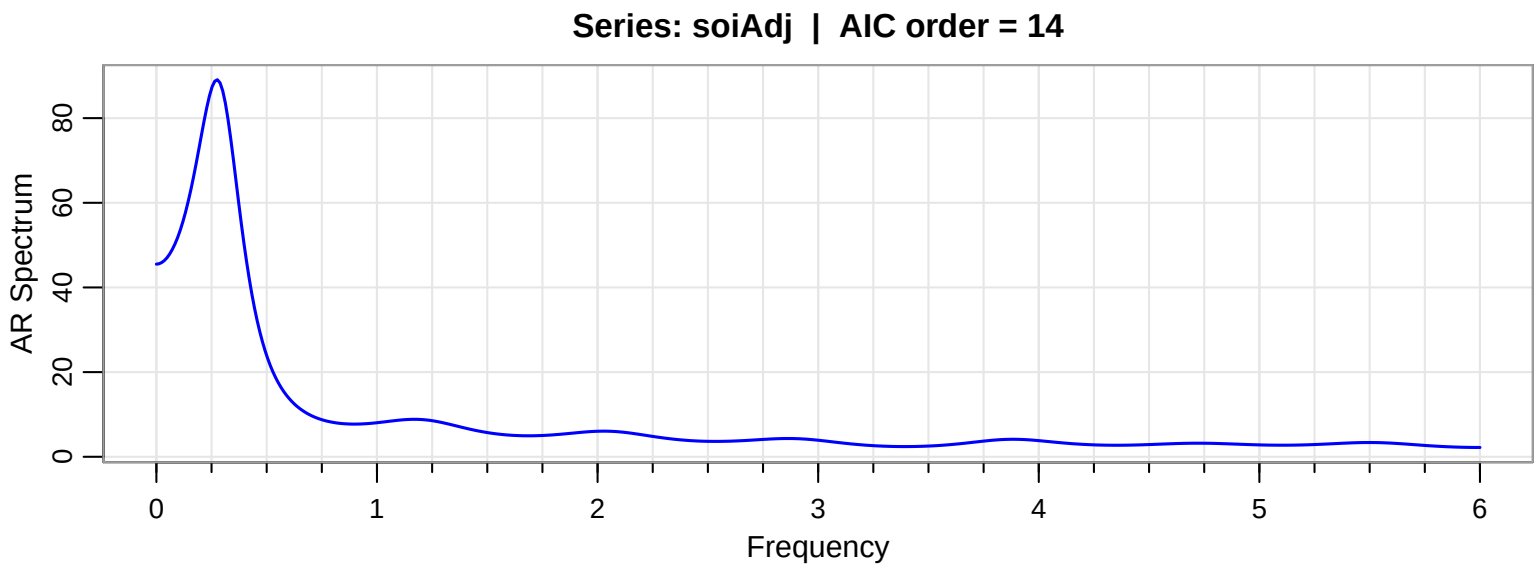
Example: AR(2) with $\phi_1 = 1.5$ and $\phi_2 = -0.95$



- Black: True
- Blue: Parametric Estimate
- Red: Non-parametric Estimate (Daniell $l = 3$)

SOI Example


Series: soi | AIC order = 15
AR Spectrum
0.00
0.10
0.20
0
1
2
3
4
5
6
Frequency

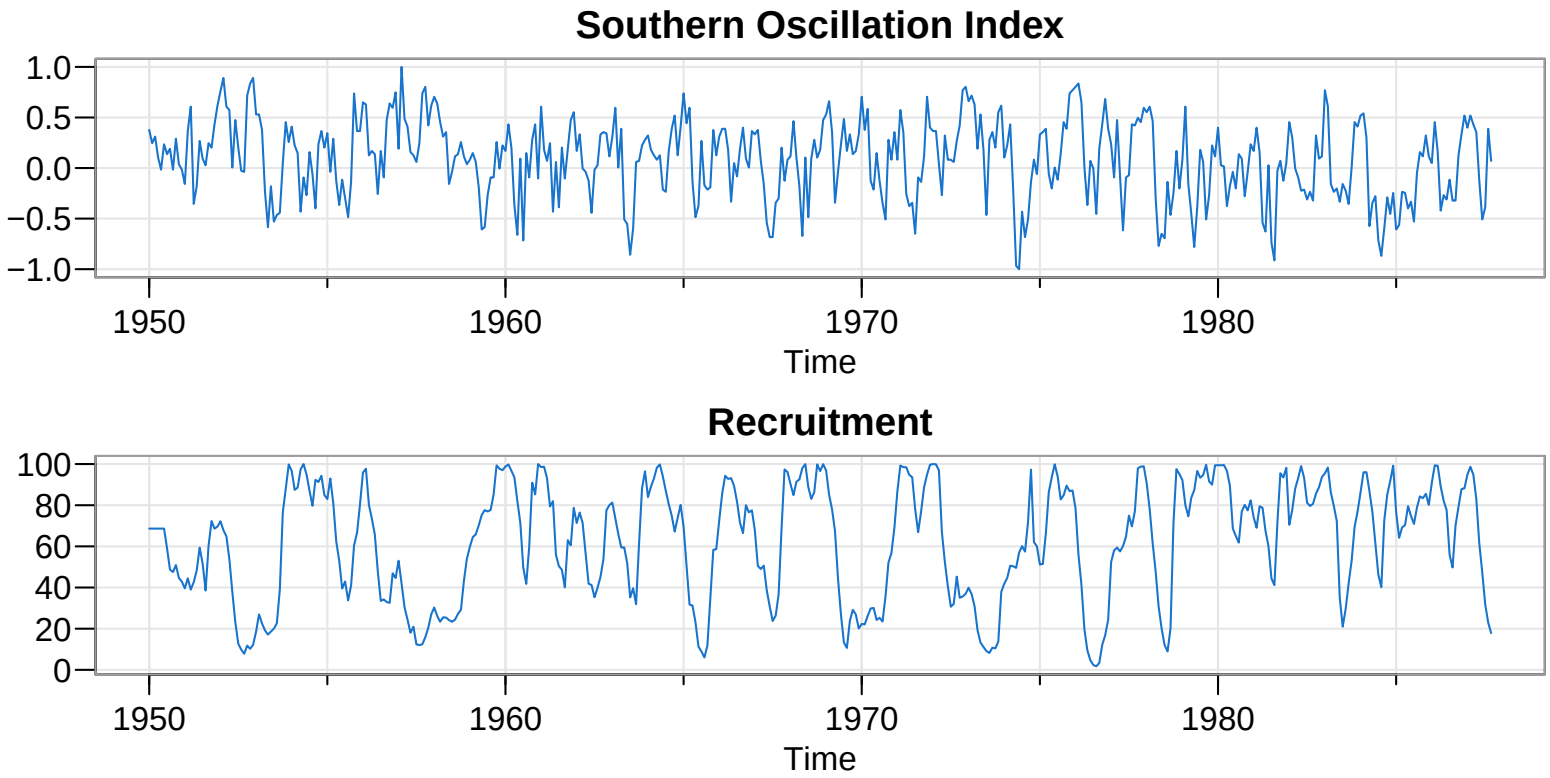

Series: soiAdj | AIC order = 14
AR Spectrum
0
20
40
60
80
0
1
2
3
4
5
6
Frequency

Why Lagged Regression?

$$Y_t - \mu_Y = \sum_{j=0}^{\infty} \beta_j (X_{t-j} - \mu_X) + V_t$$

- X_t: input time series; Y_t: output time series; V_t: error time series
- β_j: response at lag j to a unit input impulse
- **Goal**: identify delay, duration, and strength

Example: SOI and Recruitment



Input: SOI **Output**: recruitment

Question: Does SOI lead recruitment, and by how many months?

Lagged Regression: Two Complementary Views

- Time domain: CCF $\rightarrow$ prewhitening $\rightarrow$ lag response
- Frequency domain: cross-spectrum $\rightarrow$ coherence $\rightarrow$ transfer function

Both describe linear dynamic association; neither along establishes causality

Cross-Correlation: Fix the Lag Convention

The cross-covariance function of two jointly stationary processes $\{X_t\}$ and $\{Y_t\}$ is

$$\gamma_{XY}(h) = \mathbb{E}\left[(X_{t+h} - \mu_X)(Y_t - \mu_Y)\right].$$

The cross-correlation function (CCF) is

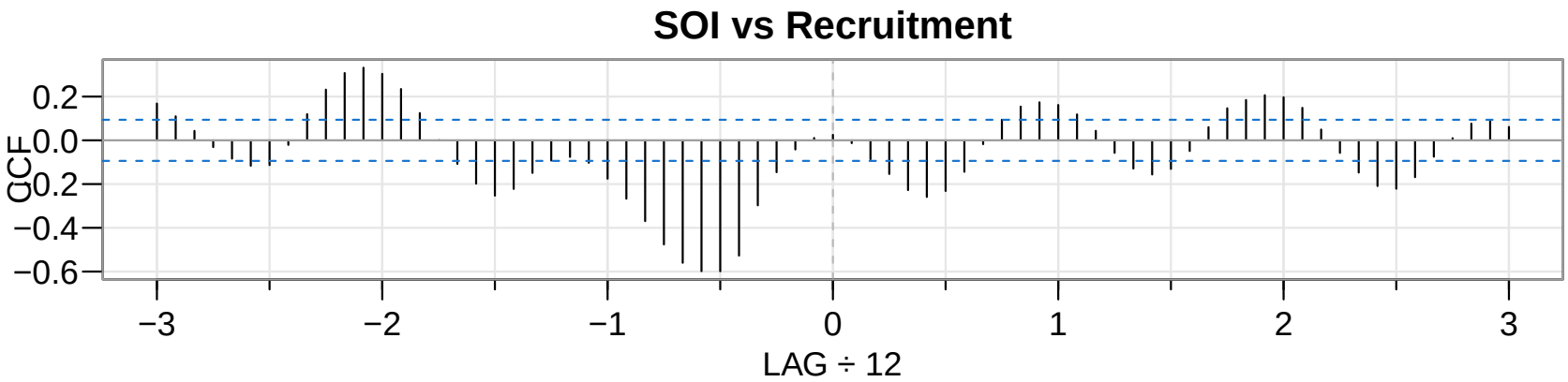
$$\rho_{XY}(h) = \frac{\gamma_{XY}(h)}{\sqrt{\gamma_X(0)\gamma_Y(0)}}.$$

For $Y_t = \beta X_{t-\ell} + V_t$, with V_t uncorrelated with X at all lags,

$$\gamma_{XY}(h) = \beta\gamma_X(h + \ell).$$

Lag convention: A negative CCF lag indicates that X leads Y.

SOI-Recruitment: What Does the CCF Show?



CCF of SOI and recruitment has a peak at $h = -6$. Thus, SOI leads recruitment by 6 months

Why can the Raw CCF Mislead?

$$Y_t = \beta X_{t-\ell} + V_t \Rightarrow \gamma_{XY}(h) = \beta\gamma X(h+\ell)$$

- Autocorrelated input $\Rightarrow$ broad CCF peaks
- CCF reflects both input persistent and response delay
- **Prewhitening** isolates the lag structure more clearly

Prewhitening: Filter Both Series

Step 1: Fit an ARMA model to the input X_t:

$$X_t = \frac{\theta_X(B)}{\phi_X(B)} W_t, \qquad \underbrace{\frac{\phi_X(B)}{\theta_X(B)}}_{A(B)} X_t = W_t.$$

Step 2: Apply the same filter to the output Y_t:

$$Y_t = \beta(B) X_t + V_t$$

$$\underbrace{A(B) Y_t}_{\tilde{Y}_t} = \beta(B) W_t + A(B) V_t.$$

Step 3: Examine the CCF of Z_t and $\tilde{Y}_t$ to identify potential input-output lags.

Key principle: Estimate the filter from X_t, then apply it to **both series** (after centering)

Prewhitened CCF Reveals the Response

After prewhitening, let

$$\tilde{Y}_t = \sum_{j=0}^{\infty} \beta_j W_{t-j} + \tilde{V}_t,$$

where W_t is white noise with variance σ_W^2.

If $\tilde{V}_t$ is uncorrelated with W_t at all lags,

$$\boxed{\gamma_{W,\tilde{Y}}(-j) = \mathrm{Cov}(W_t, \tilde{Y}_{t+j}) = \sigma_W^2 \beta_j}$$

- **Lag $-j$:** Input W_t precedes the response $\tilde{Y}_{t+j}$ by j
- **CCF magnitude:** Proportional to the response coefficient β_j
- **Normalized CCF:** Reveals the response pattern, not the coefficients directly

Key insight: Prewhitening makes the CCF proportional to the input–output impulse response.

Time-Domain Identification: Workflow

1. **Model the input:** Fit an ARMA model to X_t

2. **Prewhiten:** Apply the same filter to X_t and Y_t; examine their filtered CCF

3. **Identify the response:** Fit a parsimonious transfer-function model with appropriate delays

4. **Validate the model:** Model residual autocorrelation and check residual diagnostics

Parsimonious Transfer Functions

A transfer function summarizes the delayed response of Y_t to changes in X_t:

$$Y_t = \beta(B)X_t + V_t, \qquad \boxed{\beta(B) = B^d \frac{\delta(B)}{\omega(B)}}.$$

- d: **Delay** before the response begins
- $\delta(B)$: **Numerator** controlling the response shape
- $\omega(B)$: **Denominator** controlling persistence and decay

Example:

$$\beta(B) = \frac{0.5B^2}{1 - 0.6B} = 0.5B^2 + 0.3B^3 + 0.18B^4 + \cdots$$

Interpretation: The response begins after 2 periods and then decays geometrically

From Cross-Covariance to Cross-Spectrum

Time domain: The cross-covariance $\gamma_{XY}(h)$ describes dependence across lags

Frequency domain: Its Fourier transform gives the cross-spectral density:

$$\boxed{f_{XY}(\omega) = \sum_{h=-\infty}^{\infty} \gamma_{XY}(h) e^{-2\pi i \omega h}}$$

Unlike an ordinary spectrum, the cross-spectrum is generally **complex-valued**:

$$f_{XY}(\omega) = |f_{XY}(\omega)| e^{i\phi_{XY}(\omega)}, \qquad f_{YX}(\omega) = \overline{f_{XY}(\omega)}.$$

- **Magnitude:** Strength of cross-frequency association.
- **Phase:** Relative timing of frequency components.

Key idea: The cross-spectrum describes how two series are related at different frequencies.

Squared Coherence: Frequency-Specific R^2

The cross-spectrum measures frequency-specific association, but its magnitude depends on the scales of X_t and Y_t.

The squared coherence normalizes the cross-spectrum:

$$C_{XY}(\omega) = \frac{|f_{XY}(\omega)|^2}{f_X(\omega) f_Y(\omega)}, \qquad 0 \leq C_{XY}(\omega) \leq 1.$$

- **Near 1:** Strong linear association at frequency ω.
- **Near 0:** Weak linear association at frequency ω.
- **Analogy:** Squared coherence plays a role similar to R^2 in regression.

Estimating Coherence: Smoothing Is Essential

As in **spectral density estimation**, we smooth periodogram ordinates across nearby frequencies to obtain stable estimates

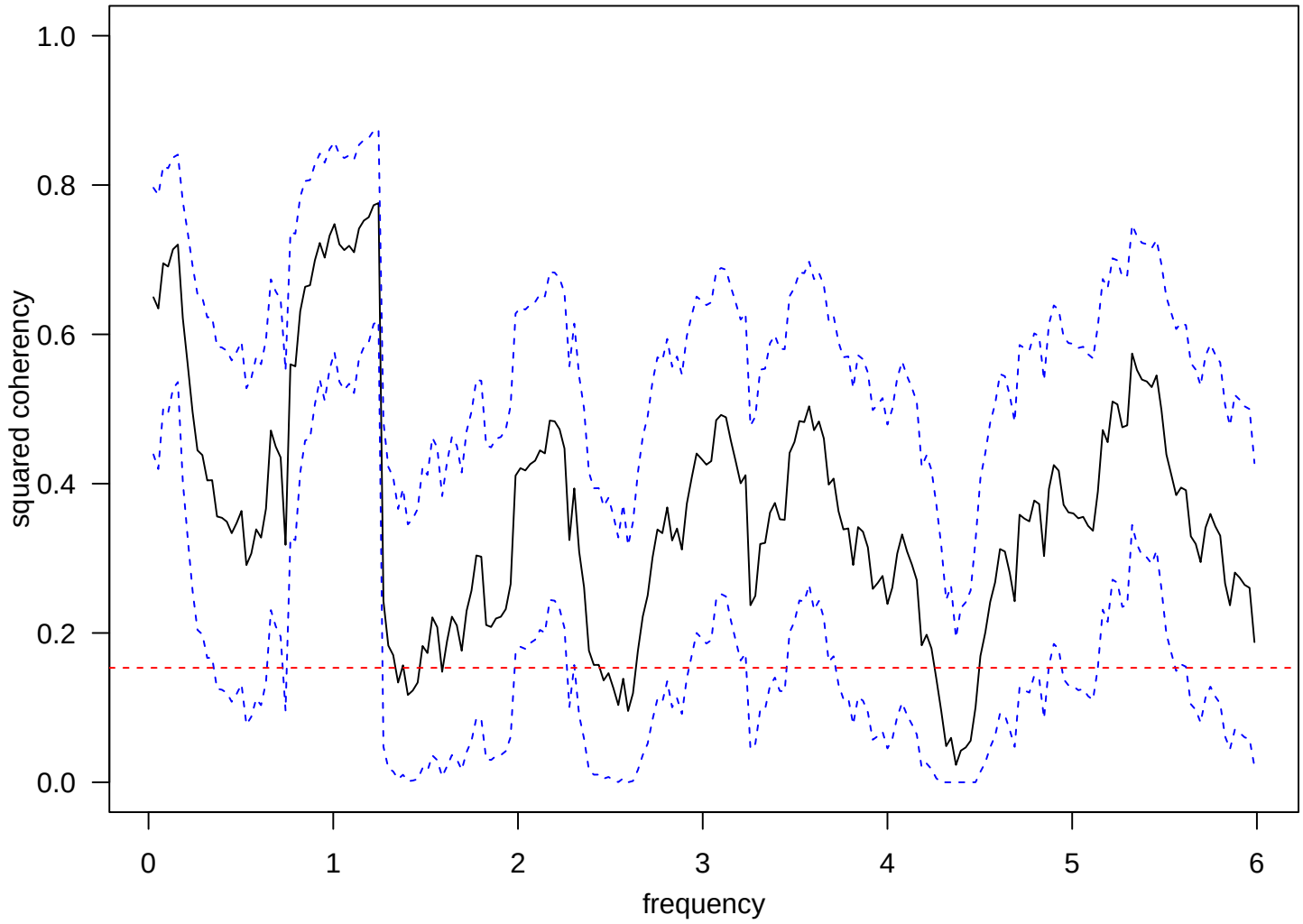
Smoothed cross-spectrum:

$$\hat{f}_{XY}(\omega_j) = \sum_k w_k \, d_X(\omega_{j+k}) \overline{d_Y(\omega_{j+k})}.$$

Estimated squared coherence:

$$\hat{C}_{XY}(\omega_j) = \frac{|\hat{f}_{XY}(\omega_j)|^2}{\hat{f}_X(\omega_j)\hat{f}_Y(\omega_j)}.$$

SOIRecruitment: Squared Coherence



Question: At which frequencies is their linear association strongest?

Two-Sided Filtering vs. Causal Lagged Regression

Unrestricted two-sided linear projection:

$$\widehat{Y}_t = \sum_{j=-\infty}^{\infty} \beta_j X_{t-j}.$$

Its normal equations yield the frequency-domain ratio

$$B(\omega) = \frac{f_{YX}(\omega)}{f_X(\omega)} \quad \text{where } f_X(\omega) > 0.$$

Causal lagged regression: restrict $\beta_j = 0$ for $j < 0$ so the output uses only current and past inputs.

The simple spectral ratio solves the unrestricted two-sided problem; a causal filter generally requires spectral factorization or constrained normal equations.

Transfer Function: From Convolution to Multiplication

Recall the two-sided linear projection:

$$Y_t = \sum_{j=-\infty}^{\infty} \beta_j X_{t-j} + V_t, \qquad \mathrm{Cov}(V_t, X_s) = 0 \quad \forall s.$$

Time domain: Cross-covariance is a convolution:

$$\gamma_{YX}(k) = \sum_{j=-\infty}^{\infty} \beta_j \gamma_X(k-j).$$

Frequency domain: Convolution becomes multiplication:

$$\boxed{f_{YX}(\omega) = H(\omega) f_X(\omega), \qquad H(\omega) = \sum_{j=-\infty}^{\infty} \beta_j e^{-2\pi i \omega j}.}$$

Key insight: The transfer function $H(\omega)$ describes how the input is transformed at each frequency.

Recovering the Impulse Response

From $f_{YX}(\omega) = H(\omega) f_X(\omega)$, we obtain the transfer function:

$$\boxed{H(\omega) = \frac{f_{YX}(\omega)}{f_X(\omega)}, \qquad f_X(\omega) > 0.}$$

The **inverse Fourier transform** recovers the impulse-response coefficients:

$$\boxed{\beta_j = \int_{-1/2}^{1/2} H(\omega) e^{2\pi i \omega_j} \, d\omega, \qquad j \in \mathbb{Z}.}$$

- **Estimate:** Smooth the auto- and cross-spectra; stabilize the spectral ratio.
- **Invert:** Recover the two-sided impulse response $\{\beta_j\}$.

Coherence and Unexplained Variation

Spectral decomposition:

$$f_Y(\omega) = \underbrace{|H(\omega)|^2 f_X(\omega)}_{\text{Explained}} + \underbrace{f_V(\omega)}_{\text{Unexplained}}$$

Using squared coherence,

$$f_V(\omega) = f_Y(\omega)\big[1 - C_{XY}(\omega)\big].$$

The minimum mean squared error is

$$\text{MSE} = \int_{-1/2}^{1/2} f_V(\omega)\, d\omega.$$

Key insight: Squared coherence measures the fraction of spectral variation explained by the optimal two-sided linear projection.

Key Takeaways

1. **Raw CCF can be misleading:** Input autocorrelation may obscure the underlying lag structure.

2. **Prewhitening reveals delays:** Filtering both series helps identify the dynamic input–output response.

3. **Coherence measures association:** Quantifies frequency-specific linear dependence, but not direction or causality.

4. **Spectral methods recover responses:** The spectral ratio identifies the optimal two-sided linear filter, not necessarily the optimal causal predictor.

Bottom line: Time-domain and frequency-domain methods provide complementary views of dynamic relationships between time series.